THE UNIVERSITY OF HONG KONG
DEPARTMENT OF MATHEMATICS

MATH2241 Introduction to Mathematical Analysis
Test 2
13 November 2020

Please write down your name and university number. Precise and adequate explanations should be given to each problem.

1. Determine whether each of the following series is convergent or divergent.

   (a) (3 points) $\displaystyle\sum_{n=1}^{\infty} \frac{\tan(\sin n)}{n^3}$

   (b) (4 points) $\displaystyle\sum_{n=1}^{\infty} \frac{1}{\ln(2n^n+1)}$

2. (7 points) Define $f:\mathbb{R}\to\mathbb{R}$ by

$$f(x)=\begin{cases} x^2-1 & \text{if } x\in\mathbb{Q},\\ -\dfrac{(x+1)(3x+1)}{x} & \text{if } x\notin\mathbb{Q}.\end{cases}$$

   Find all real numbers a such that f is continuous at a.

3. (7 points) Define $g:\mathbb{R}\to\mathbb{R}$ by $g(x)=\dfrac{2x+1}{x^2+2}$. Prove that g is uniformly continuous.

4. (7 points) Let $f:[a,b]\to\mathbb{R}$ be a continuous function defined on a closed and bounded interval. It is known that for any $x\in(a,b)$, for any $\delta>0$, there exists $y\in[a,b]\cap(x,x+\delta)$ such that $f(y)=f(x)$. Prove that f is a constant function.

5. Prove or disprove each of the following. State clearly whether it is true or false.

   (a) (4 points) Let $\{a_n\}$ be a sequence of positive real numbers. If the series $\displaystyle\sum_{n=1}^{\infty}\sqrt{a_n}$ diverges, then the series $\displaystyle\sum_{n=1}^{\infty}\sqrt{\frac{a_n^2+1}{2a_n}}$ diverges.

   (b) (4 points) Let $\{b_n\}$ be a sequence of nonnegative real numbers. Then the series $\displaystyle\sum_{n=1}^{\infty} b_n$ converges if and only if the series $\displaystyle\sum_{k=1}^{\infty} b_{n_k}$ converges for any subsequence $\{b_{n_k}\}$ of $\{b_n\}$.

   (c) (4 points) Let $f:\mathbb{R}\to\mathbb{R}$ be a function. If the series $\displaystyle\sum_{n=1}^{\infty} f\left(\frac{1}{n}\right)$ converges, then $\lim_{x\to 0} f(x)=0$.